Multi-Sleeve Statistical Arbitrage Portfolio Construction for U.S. Equities

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1 Introduction

This paper explores the construction of a diversified multi-sleeve statistical arbitrage portfolio using S&P 500 equities. The portfolio combines six signal families including momentum, active return mean reversion, residual mean reversion, distance pairs relative value, sector relative mean reversion, and event driven gap reversion into a dollar neutral long short framework. I use regime gating, risk scaling, and iterative portfolio construction to combine low correlation sleeves into a diversified portfolio robust across weighting schemes and ablation tests. The final portfolio achieves an in-sample net Sharpe ratio above 2.0 after 10 basis points of transaction costs during the 2015–2023 period. During the out-of-sample period (2024–2026), performance weakens but remains profitable while preserving much of the portfolio's diversification structure. The results suggest that diversification across complementary momentum and mean reversion sleeves may produce more stable portfolio level behavior than reliance on any single standalone signal [1, 2].

2 Methodology & Experimental Setup

All data is sourced from yfinance daily close prices. The universe consists of S&P 500 equities filtered each day to the 300 most liquid names using 60 day rolling dollar volume. SPY is used as the benchmark and market factor, while sector exposures are modeled using the major sector ETFs (XLK, XLF, XLE, XLV, XLI, XLY, XLP, XLU, XLRE, XLB). Signals are developed on an in-sample period from 2015–2023 and evaluated out-of-sample from January 2024 through April 2026.

Portfolio construction uses a dollar neutral long short framework with 20 long and 20 short positions equally weighted within each sleeve. Signals are rank transformed cross sectionally and shifted forward one trading day to eliminate lookahead bias. Rebalancing frequencies of 1, 5, 10, and 21 trading days are tested depending on signal horizon. Transaction costs are modeled as a constant 10 basis point linear cost per trade with no explicit market impact assumptions. All sleeves also use an equity curve risk scaler that reduces exposure to 25% of nominal size whenever the sleeve equity curve falls below its 20 day moving average.

The research process moves from broad signal discovery toward progressively more structured portfolio construction and robustness testing. I first explore multiple variants of momentum, mean reversion, distance pairs, sector relative, and event driven signals using different return definitions, lookback windows, and rebalance frequencies while deliberately avoiding aggressive parameter tuning. The aim is to identify signal families exhibiting persistent predictive structure without relying on aggressive tuning. The strongest signals are then rebuilt with regime conditioning and risk scaling overlays using filters based on trend, volatility, breadth, dispersion, panic regimes, and sector dislocations. After conditioning, I analyze return correlations between sleeves to identify complementary combinations and incrementally construct larger portfolios from low correlation components.

Finally, I test the robustness of the portfolio architecture across different investment styles and weighting schemes including equal weight, volatility scaled, Sharpe scaled, and ridge regularized mean variance optimization. I also perform sleeve and family ablation tests to verify that performance is driven by diversification across complementary sleeves rather than dependence on any single signal family.

3 Signal Discovery

I test six different signal families: momentum, active return mean reversion, residual return mean reversion, sector-relative mean reversion, distance-pairs mean reversion, and gap reversion [1, 2]. Table 1 gives the definitions and economic intuition behind each signal.

Signal Family	Formula	Description
Monotonic Momentum	$h_{i,t}^{(w)} = \frac{1}{w}\sum_k \mathbf{1}(\text{sign}(r_{i,t-k}) = \text{sign}(\mu_{i,t-k}^{(w)}))$ $s_{i,t}^{\text{mon}} = h_{i,t}^{(w)} \cdot \lvert\mu_{i,t}^{(w)}\rvert$	Rewards path consistency, not just cumulative return. Multiplies the fraction of days where the return agreed in sign with the rolling mean by the magnitude of that mean. Stocks trending persistently in one direction, not just on average, are more likely to continue.
Active-Return MR	$s_{i,t}^{\text{mr}} = -\bar{r}_{i,t}^{(w)}$ $s_{i,t}^{\text{zrev}} = -\frac{\bar{r}_{i,t}^{(f)} - \mu_{i,t}^{(s)}}{\sigma_{i,t}^{(s)}}$	Bets that stocks which recently outperformed or underperformed the market tend to revert. The plain variant negates the rolling active return. The z-score variant standardizes the short-term deviation from a long-term baseline by the stock's own active-return volatility. The same raw move counts less when the stock is already volatile.
Residual MR	$s_{i,t}^{\text{res}} = -\bar{\tilde{r}}_{i,t}^{(w)}$	Applies the same reversal logic to returns residualized against market beta and sector factors. Removing common factor exposures isolates the idiosyncratic moves preventing market or sector moves from contaminating the cross-sectional signal.
Distance-Pairs MR	$z_{i,t} = \frac{d_{i,t} - \bar{d}_{i,t}^{(w)}}{\sigma^{(w)}(d_{i,t})}$ $s_{i,t}^{\text{pairs}} = -\,\text{clip}(z_{i,t}, -2, 2)$	Trades relative-value convergence among historically similar stocks. For each stock, identifies k nearest partners using a correlation-based distance metric on normalized log price paths. The signal is the clipped z-score of the deviation from the average partner path.
Sector-Relative MR	$s_{i,t}^{\text{sector}} = -\overline{(r_{i,t} - r_{g(i),t}^{\text{sector}})}^{(w)}$	Fades moves relative to the stock's sector ETF rather than the broad market, where $g(i)$ maps each stock to its sector. Isolates within-sector dislocations that may revert as the stock converges back toward sector peers.
Gap Reversion	$s_{i,t}^{\text{gap}} = -\tilde{r}_{i,t-1}$	Fades large single-day idiosyncratic price shocks. A large positive residual move yesterday produces a negative signal today. Uses factor-neutral returns to ensure the effect is in the idiosyncratic component and not in market or sector moves.

Table 1: Signal family definitions.

The base rows of Table 2 show each signal family's performance before any conditioning or risk scaling. The monotonic momentum signal uses a 120-day window. When I removed the magnitude weighting, performance plummeted ($0.606 \rightarrow 0.177$). This suggests the signal's strength comes from path consistency: a stock that drifts upward on most days is more likely to continue than one with the same cumulative return concentrated in a few large moves.

For active-return mean reversion, I use a z-score variant that standardizes each stock's short-term active return deviation by its own volatility. For residual mean reversion, I test both a cross-sectional factor-model residualization and a time-series ETF-factor residualization approach. The ETF-factor branch produces the stronger standalone signal (0.694 vs 0.179 base Sharpe), but the factor-model variant responds better to conditioning because it isolates idiosyncratic dislocations more aggressively.

Distance-pairs mean reversion identifies tight relative-value relationships that revert reliably, but the signal updates frequently as z-scores shift producing higher turnover and transaction costs. Sector-relative mean reversion starts weak suggesting that within-sector dislocations are noisier than market-wide or idiosyncratic ones and require additional filtering to become tradable. Gap reversion isolates single-day residual shocks and captures overnight repricing of idiosyncratic moves, which explains both its moderate standalone strength and its relatively low correlation with the other signal families.

Several patterns stand out when gating signals to market regimes (see Table 2). Both the active-return and residual mean reversion variants achieve their highest net Sharpes when gated to high residual-dispersion regimes, periods when cross-sectional idiosyncratic movement is elevated and reversal signals have more dislocations to work with. The active-return z-score variant responds especially well to filtering, reaching a 1.279 net Sharpe, the highest conditioned signal across all families. The factor-model residual variant also responds more strongly to filtering than the ETF-factor variant despite its weaker standalone performance. Sector-relative mean reversion improves substantially under filtering (0.227 → 0.886). By contrast, distance-pairs shows almost no improvement from filtering. The raw relative-value convergence signal is already strong on its own because the partner-matching mechanism implicitly conditions on the stability of the peer relationship.

Risk scaling tells a similar story (see Table 2, scaled rows). Uptrend scaling substantially improves the mean reversion families while cutting turnover by more than half. The intuition is that mean reversion signals work best when dislocations are temporary rather than part of a persistent repricing process. Momentum responds better to a low-dispersion scaler, which reduces exposure when cross-sectional dispersion becomes elevated and stock paths become noisier. Gap reversion benefits from a combination of uptrend scaling and volatility-contraction gating, pushing net Sharpe to 0.825 while reducing turnover from 11.1% to 3.6%.

Overall, conditioning and risk scaling improve implementation quality primarily by reducing turnover and avoiding noisy regimes rather than by increasing gross alpha directly.

Family	Variant	Net SR	Gross SR	Turnover	Max DD	ρ_{bench}
Monotonic Momentum	Base (120d, rebal 21d)	0.606	0.763	5.072%	−14.708%	0.262
	+ low-dispersion scaling (60d, q30)	0.769	0.910	3.983%	−10.146%	0.253
Active-Return MR	Base (z-score 5d/60d, rebal 10d)	0.372	0.834	11.490%	−11.741%	0.189
	+ uptrend scaling (20/100d)	0.644	0.908	4.792%	−6.069%	0.160
	+ high residual dispersion gate (20d, q75)	1.279	1.426	1.733%	−3.173%	0.143
Residual MR	Base (factor-model 5d, rebal 10d)	0.179	0.687	10.946%	−9.197%	0.145
	+ uptrend scaling (20/100d)	0.695	1.034	4.945%	−4.227%	0.140
	+ high residual dispersion gate (20d, q75)	1.083	1.192	1.542%	−4.538%	0.092
Distance-Pairs MR	Base (k=3, 60d z-score, rebal 10d)	1.078	1.549	12.911%	−9.709%	−0.179
Sector-Relative	Base (sector-rel. 5d, rebal 10d)	0.227	0.694	11.999%	−15.969%	0.147
	+ uptrend scaling (20/100d)	0.551	0.848	5.339%	−8.177%	0.134
	+ high residual dispersion gate (20d, q75)	0.886	1.014	2.021%	−5.926%	0.019
Event / Gap Reversion	Base (residual gap reversal, rebal 10d)	0.626	1.178	11.113%	−10.508%	0.128
	+ uptrend scaling (50/200d) + vol contraction gate	0.825	1.093	3.599%	−4.383%	0.043

Table 2: Signal conditioning progression from base to best risk-scaled and conditioned variants.

4 Portfolio Construction

4.1 Sleeve Correlations

I first look at return correlations across conditioned and risk-scaled sleeves to understand which families might pair well together and why. Figure 1 reveals two natural clusters. Active-return MR, residual MR, and sector-relative MR form a correlated block with pairwise correlations in the 0.37 to 0.56 range. This makes economic sense since these are all expressions of the same underlying cross-sectional reversal phenomenon.

Residualization and sector anchoring change the implementation characteristics and regime behavior, but the core bet is similar. Stocks that moved too much recently should revert.

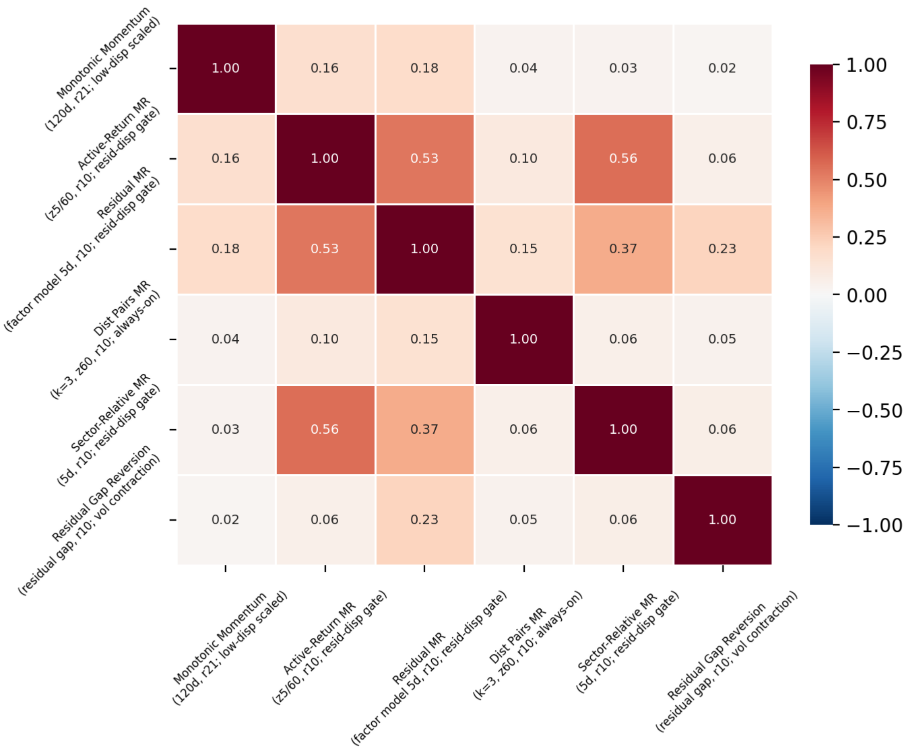

Figure 1: Return correlations across conditioned sleeves. Two natural groups emerge: a correlated mean-reversion complex and a set of orthogonal diversifiers.

Distance-pairs mean reversion (DPMR) stands apart from this cluster. Its correlations to every other family stay around 0.04 to 0.15. Even though DPMR is also a form of mean reversion, its reference point is different. It trades convergence toward a stock's nearest historical peers rather than reversal toward a cross-sectional or factor-adjusted mean. That distinction produces a return stream that behaves like a separate alpha source.

Monotonic momentum correlates only 0.16 to 0.18 with the MR families. This is low enough to suggest that momentum and reversal are complementary rather than offsetting. A momentum sleeve paired with reversal sleeves should produce diversification rather than cancellation.

Event-driven gap reversion also appears fairly independent with correlations mostly below 0.23. It captures short-horizon idiosyncratic dislocations on a timescale distinct from both the slower reversal processes and the multi-week momentum signal.

No pair of families exhibits dangerously high correlation. Even the strongest relationship (active-return MR and sector-relative MR at 0.56) is moderate rather than redundant. This suggests that a multi-engine portfolio combining sleeves across these clusters should produce materially better risk-adjusted performance than selecting any single best signal.

Notably, conditioning doesn't collapse the sleeves into the same regime trade. Despite many sleeves using similar filters (dispersion, breadth, volatility contraction), the realized return streams remain differentiated. Conditioning should help improve the implementation without destroying the diversification that makes portfolio economically appealing.

4.2 Sleeve Selection

The two strongest standalone signals after conditioning are the mean reversion sleeves: active-return MR (1.279 net Sharpe) and residual MR (1.083 net Sharpe). I begin the portfolio buildout by combining these two sleeves despite their moderate correlation ($\sim$0.4), then look for additional sleeves that can improve not only Sharpe but also drawdown and turnover characteristics. The portfolio is built incrementally by adding one sleeve at a time and tracking the marginal impact on each metric. Table 3 summarizes the sleeve slection process.

Step	Sleeve Added	Sleeves	Net SR	Gross SR	Turnover	Max DD
1	Residual MR (high resid. disp. gate)	1	1.074	1.181	1.542%	−4.526%
2	+ Active-Return MR (breadth gate)	2	1.242	1.339	1.655%	−6.020%
3	+ DPMR (z20, vol contraction gate)	3	1.556	1.746	2.447%	−3.753%
4	+ DPMR (z60, vol expansion gate)	4	1.720	2.031	3.799%	−3.105%
5	+ DPMR (z60, always-on)	5	1.786	2.219	5.536%	−3.254%
6	+ DPMR (z10, panic gate)	6	1.883	2.297	4.699%	−2.811%
7	+ Monotonic Momentum (crash-gated)	7	1.989	2.417	4.619%	−3.196%
8	+ Resid. Gap Reversion (breadth-gated)	8	2.039	2.548	5.020%	−2.347%

Table 3: Incremental portfolio buildout. Each row adds one sleeve and reports aggregate statistics.

The portfolio starts with the two strongest sleeves from the correlated MR cluster. Combining them improves Sharpe from 1.07 to 1.24, but because they share moderate correlation, the combination doesn't dramatically reduce drawdown. Diversification of the portfolio really starts when adding the distance-pairs sleeves. Sharpe jumps from 1.24 to 1.56 while max drawdown also drops from −6.0% to −3.8%. The low correlation of DPMR to the MR sleeves drives the improvement. As discussed in the correlation analysis, DPMR trades convergence toward historical peers rather than cross-sectional reversal, producing a genuinely distinct return stream despite also being a form of mean reversion.

Additional DPMR variants (see Table 4) continue adding incremental value even though they belong to the same signal family. All four DPMR sleeves share the same base construction ($k = 3$ nearest partners, z-scored spread reversal clipped to $[-2, 2]$), but they differ along three key dimensions that make them economically distinct.

Sleeve	Z Window	Rebal.	Conditioning	Role
DPMR z10 + panic	10	10d	Panic regime (−5% in 10d)	Fast crisis-reversion; episodic
DPMR z20 + vol contr.	20	21d	Vol contraction (10d < 60d vol)	Slow calmer market
DPMR z60 + vol exp.	60	5d	Vol expansion (10d > 60d vol)	Dislocation in turbulence
DPMR z60 + none	60	10d	Always on	Baseline structural convergence

Table 4: Distance-pairs sleeve parametrization.

The conditioning filters effectively partition the same alpha engine across different market environments rather than creating duplicate exposure. The panic sleeve (z10) is the most reactive. It only trades after severe market drawdowns and has extremely low realized turnover (0.6% daily) because it's rarely active. It functions as an episodic crisis-reversion bet with minimal drag in normal environments. The vol-contraction sleeve (z20) activates during calmer regimes where pair spreads are steadily reverting. The vol-expansion sleeve (z60) captures persistent dislocations in turbulent environments using the fastest rebalance (5d). The always-on sleeve (z60) is the structural convergence engine. Since it has no gate, it trades in all regimes which leads to high turnover (12.9% daily), but also the highest standalone Sharpe (1.549).

Together these four sleeves form the portfolio's core diversification. Adding all four DPMR variants (steps 3 to 6), the portfolio reaches a 1.883 Sharpe with a −2.8% max drawdown. The always-on sleeve notably increases turnover (3.8% → 5.5% at step 5), but the panic sleeve at step 6 reduces aggregate turnover back to 4.7%. Its episodic nature means it contributes alpha without adding continuous trading burden.

With the core DPMR and MR sleeves in place, I tested additional sleeves from the remaining uncorrelated families — monotonic momentum, event-driven gap reversion, and sector-relative mean reversion — to further diversify the portfolio and improve its properties beyond Sharpe alone. These families operate on different timescales and capture different types of dislocations, so the goal was to find sleeves that could reduce drawdown or smooth returns rather than simply add more of the same alpha.

Monotonic momentum (crash-gated) improves Sharpe from 1.88 to 1.99 while drawdown stays controlled. The final residual-gap reversion sleeve (breadth-gated) pushes Sharpe to 2.04 while simultaneously reducing max drawdown to −2.35%, improving both return efficiency and downside behavior. Sector-relative sleeves were also tested and contributed modestly as secondary diversifiers but were not included in the final portfolio because they added less incremental value than the momentum and gap reversion alternatives.

The net result is an 8-sleeve portfolio with nearly double the Sharpe of the best standalone signal. Every sleeve addition improves the portfolio monotonically across all three key metrics.

4.3 Robustness

I test the portfolio's stability along two dimensions: sensitivity to the weighting scheme and sensitivity to sleeve removal (ablation).

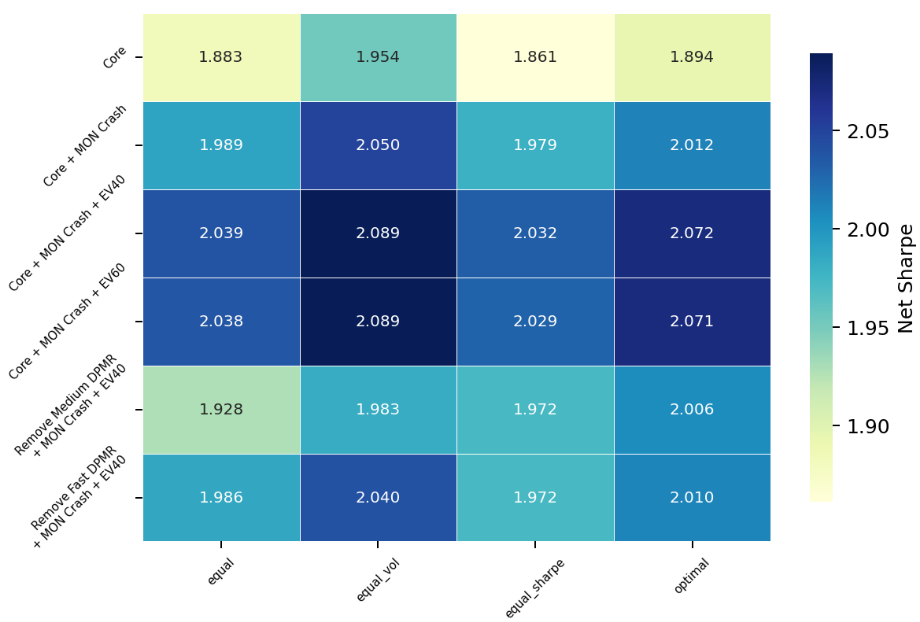

Figure 2: Sharpe ratio across extension variants and weighting schemes. The portfolio remains above 1.8 Sharpe across a wide range of configurations.

The portfolio is robust across weighting schemes. Equal-weight, equal-volatility, equal-Sharpe, and ridge-regularized mean-variance optimization all produced net Sharpes above 1.8. This indicates that the diversification benefit is structural rather than dependent on getting the weights exactly right.

Removing individual sleeves caused limited Sharpe degradation. No single sleeve removal dropped the portfolio below 1.8 Sharpe, confirming that the portfolio's strength comes from the diversification structure itself rather than from any one dominant alpha engine. The portfolio can tolerate the loss of any individual component without collapsing.

The portfolio is also insensitive to specific parameterization choices within sleeves. Varying the breadth gate threshold on the event sleeve (40th vs 60th percentile) produced effectively identical results (2.039 vs 2.038 Sharpe). Adding a second event sleeve did not improve on the simpler configuration, and sector-relative sleeves contributed modestly as secondary diversifiers (1.972 Sharpe) but were not necessary for the portfolio to maintain strong performance. The key robustness finding is not the specific 2.039 Sharpe of the best configuration, but how stable the result was across nearby configurations.

4.4 Sleeve Attribution

Table 5 shows that returns are distributed relatively evenly across sleeves. The largest contributor accounts for 16.1% of returns and the smallest 9.2% and most sleeves contributing between 10% and 14%. No single sleeve dominates the portfolio, suggesting that the Sharpe comes from diversification across multiple alpha engines rather than concentration in one dominant strategy.

The attribution profile also highlights the different functional roles of the sleeves. The vol-contraction DPMR sleeve is the most efficient component contributing 13.8% of returns while accounting for only 7.3% of variance. The residual gap-reversion sleeve is the most operationally expensive component (28.5% of portfolio turnover) but contributes 12.2% of returns with only 10% of variance. It is expensive to trade, but it provides genuinely orthogonal alpha that justifies the cost. By contrast, the residual MR and active-return MR sleeves act as lower-turnover portfolio anchors that stabilize the portfolio. During the portfolio's worst drawdown window during the in-sample period (see Figure 4), several core relative-value sleeves remain positive even while the panic DPMR and momentum sleeves loses money. This result suggests that the portfolio's resilience comes from structural diversification across sleeves rather than from any dedicated hedge component.

Sleeve	Weight	Net SR	Ret. Contrib.	Var. Contrib.	Efficiency	Turn. Contrib.
DPMR z60, always-on	8.9%	1.549	16.1%	13.5%	1.189	23.0%
DPMR z60, vol expansion	10.7%	1.394	14.4%	14.1%	1.025	17.0%
DPMR z20, vol contraction	16.2%	1.336	13.8%	7.3%	1.900	13.0%
Residual MR (high resid. disp.)	15.8%	1.192	12.3%	16.4%	0.751	4.9%
Resid. Gap Rev. (breadth)	14.7%	1.177	12.2%	10.0%	1.224	28.5%
Active-Return MR (breadth)	9.4%	1.098	11.4%	16.4%	0.693	3.3%
DPMR z10, panic	15.8%	1.016	10.5%	13.0%	0.812	2.0%
Monotonic Momentum (crash)	8.4%	0.888	9.2%	9.3%	0.986	8.3%

Table 5: Sleeve attribution for the final 8-sleeve equal-volatility portfolio.

5 Implementation

5.1 Transaction Costs, Turnover, and Capacity

Aggregate portfolio turnover is moderate at 4.7% daily, but the execution burden is concentrated in a small subset of sleeves rather than spread evenly across the portfolio. Figure 3 shows the sensitivity of each sleeve and the overall portfolio to transaction costs. The highest turnover sleeves are also the most fragile standalone implementations. The residual gap reversion sleeve (9.7% daily turnover) falls from a 1.177 gross Sharpe to roughly zero at 20 bps and the always on DPMR sleeve (12.9% turnover) declines from a 1.549 gross Sharpe to 0.373 net at 25 bps.

The diversified portfolio behaves much more robustly. At the baseline 10 bps assumption, the portfolio retains a 2.089 net Sharpe despite roughly 20% cost drag. At 25 bps, the portfolio still maintains a 1.304 Sharpe even though several individual sleeves become marginal or unprofitable. This robustness comes from combining faster alpha generating sleeves with slower conditioned sleeves whose turnover remains low across all environments. The residual MR and active return MR sleeves are especially stable maintaining relatively high Sharpe ratios even under elevated costs.

Conditioning is a major reason the turnover profile remains manageable. Across several families, regime gating sharply reduces trading activity while simultaneously improving post cost Sharpe. The active return MR sleeve illustrates the effect most clearly: turnover falls from 11.7% to 1.8% after trend scaling and residual dispersion gating, while Sharpe rises from 0.557 to 1.279. Similar reductions appear in residual MR, sector relative, and gap reversion sleeves. In practice, conditioning acts as both a signal quality filter and an implementation control mechanism by avoiding trades during noisy or low alpha regimes.

The unconstrained always on DPMR sleeve is the dominant execution bottleneck in the portfolio. Its high turnover creates continuous pair spread trading activity across all market regimes, making it especially sensitive to slippage, crowding, and spread compression as capital scales. The residual gap reversion sleeve faces a different constraint because it concentrates trading immediately after large idiosyncratic moves when liquidity is weakest and spreads are widest. Overall, the portfolio appears reasonably implementable at

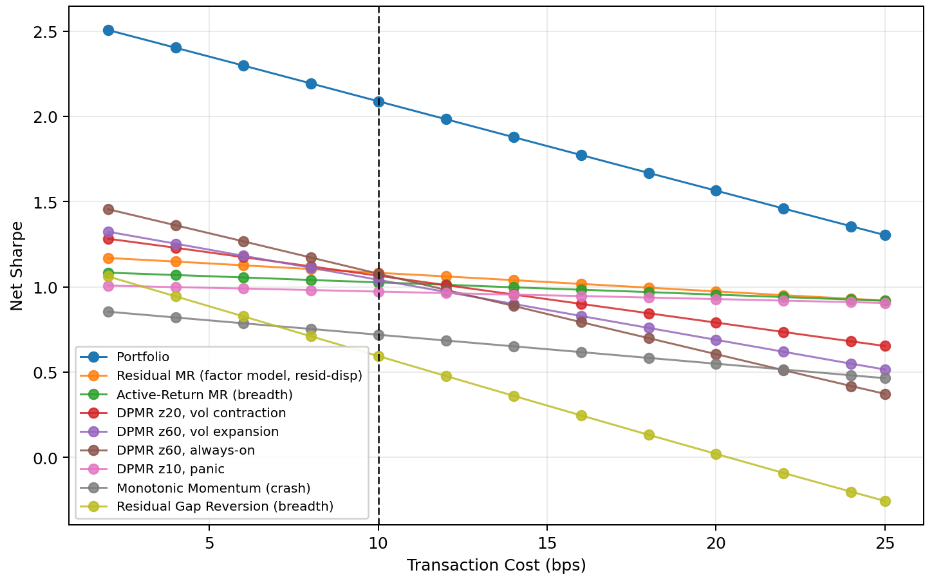

Figure 3: Portfolio and sleeve-level net Sharpe across transaction cost assumptions from 2 to 25 bps.

moderate scale because the slower conditioned sleeves offset the faster pair trading and event driven components, but the real capacity limits remain concentrated in a few short horizon sleeves rather than distributed uniformly across the portfolio.

Sleeve	Daily Turnover	Horizon	Capacity Notes
Portfolio (aggregate)	**4.700%**	**Multi-day**	**Moderate aggregate feasibility**
DPMR z10 (panic)	0.600%	Episodic	Stressed windows only; low continuous burden
Residual MR (resid. disp.)	1.500%	Medium	Easiest sleeve to scale
Active-Return MR (breadth)	1.800%	Medium	Stable core sleeve
DPMR z20 (vol contraction)	4.000%	Med-long	Monthly rebalance; requires crossing discipline
Monotonic Momentum (crash)	4.900%	Med-long	Longer holding periods; cleaner execution
DPMR z60 (vol expansion)	8.000%	Short-med	Faster rebalance; execution-dependent
Resid. Gap Rev. (breadth)	9.700%	Short	Event-driven; trading quality matters
DPMR z60 (always-on)	12.900%	Medium	Highest turnover; no conditioning filter

Table 6: Turnover and capacity profile by sleeve.

5.2 Leverage and Volatility Targeting

The final portfolio maintains low realized volatility (∼2% annualized unlevered) and controlled drawdowns, making leverage a natural scaling mechanism rather than a source of structural fragility. After conditioning and diversification, the portfolio behaves more like a diversified market-neutral risk-premia process than a directional equity strategy, so I evaluate several target-volatility overlays to convert the high risk-adjusted returns into economically meaningful absolute returns. Figure 4 shows the resulting equity curves under different target volatility assumptions. Scaling to a 10% target volatility substantially increases cumulative returns while maintaining moderate drawdowns, with maximum drawdown on the levered series remaining near 14%. Importantly, the conditioning and diversification structure remain stable under leverage: the low cross-sleeve correlations, turnover reduction from conditioning, and distinct economic roles played by the sleeves continue functioning after scaling. This stability is what makes controlled leverage feasible despite the portfolio's exposure to short-horizon sleeves.

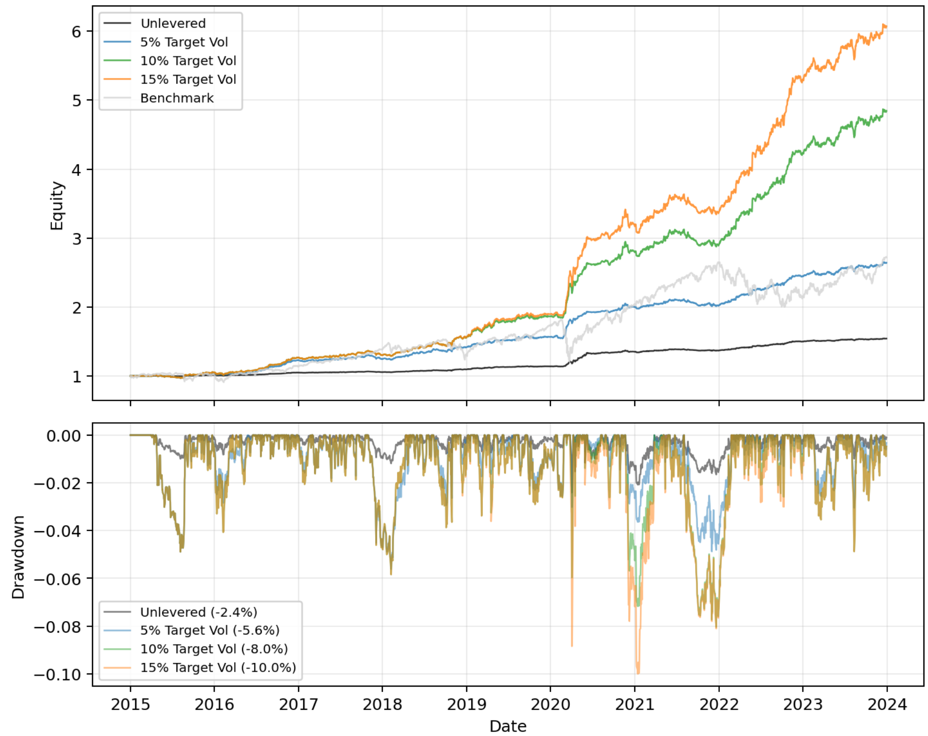

Figure 4: Equity curves at different target volatility levels.

6 Out-of-Sample Results

On the out-of-sample period, the portfolio degrades but does not collapse. As shown in Figure 5, post-2024 returns flatten, rolling Sharpe compresses, and drawdowns increase, yet the portfolio remains profitable and the equity curve continues trending upward. The behavior suggests regime sensitivity rather than obvious overfitting.

The diversification structure also persists out-of-sample. Benchmark correlation remains low, drawdowns stay materially smaller than SPY's, and realized volatility remains controlled even through the 2024 regime shift. Table 7 summarizes the annual results under 10% target-volatility scaling. The levered portfolio produced 12–20% returns in favorable environments, -9.2% in the worst year (2024), and a maximum drawdown of −11.4%. The 50.1% return in 2020 reflects cross-sectional dispersion exploding during COVID — extreme but economically consistent with the portfolio design.

2024 is the only negative year (−9.2%, −1.350 Sharpe). The market environment was dominated by strong index trends and concentrated mega-cap leadership, reducing cross-sectional reversal opportunities. A portfolio built primarily on short-horizon mean reversion, statistical arbitrage, and relative-value convergence would be expected to struggle in exactly this regime. The momentum persistence was unfavorable to the portfolio's predominant mean-reversion construction.

The recovery after 2024 is important for distinguishing regime sensitivity from overfitting. Rather than continuing to decay, the portfolio recovers in 2025 (0.840 Sharpe, +12.8%) and strengthens further in 2026 through April (2.136 Sharpe, +31.5%). This drawdown-recovery pattern supports the interpretation that the weakness was regime-driven rather than structural failure. Rolling Sharpe dynamics also remained stable under leverage, suggesting that the conditioning and diversification structure continued functioning out-of-sample.

The portfolio performs best during high-dispersion environments. The strongest years are 2020 (COVID volatility, 3.235 Sharpe) and 2022 (rate-driven selloff, 3.388 Sharpe), periods where cross-sectional opportunity expanded as stocks moved more independently. In these environments, the crisis-conditioned DPMR and event sleeves activated more frequently while several volatility-gated sleeves engaged during vol expansion.

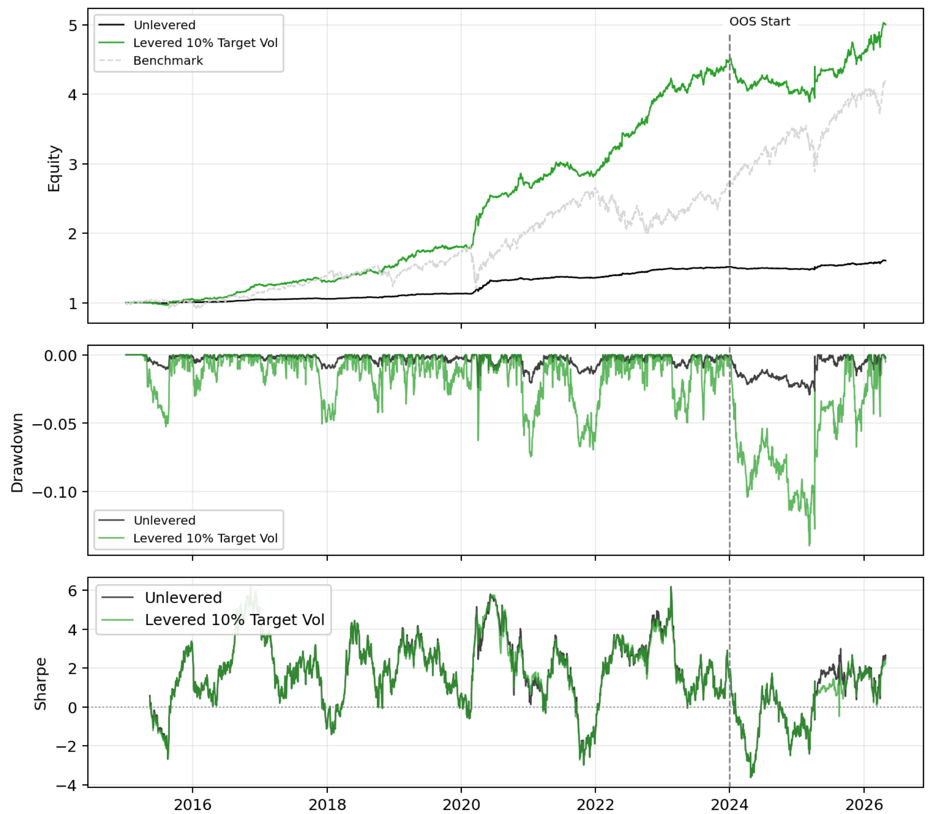

Figure 5: Portfolio equity curve from 2015 through April 2026, with the out-of-sample boundary at January 2024.

Year	Period	Lev. Ret.	Lev. SR	Lev. Vol	Lev. Max DD	SPY Ret.
2015	IS	5.400%	1.010	10.000%	−5.300%	1.300%
2016	IS	20.100%	3.064	10.000%	−2.100%	12.000%
2017	IS	2.600%	0.481	10.000%	−5.100%	21.800%
2018	IS	16.700%	2.116	10.000%	−4.200%	−4.600%
2019	IS	19.000%	2.251	10.000%	−3.600%	31.200%
2020	IS	50.100%	3.235	10.000%	−6.300%	18.300%
2021	IS	4.200%	0.528	10.000%	−7.000%	28.700%
2022	IS	40.900%	3.388	10.000%	−2.600%	−18.200%
2023	IS	12.000%	1.466	10.000%	−5.000%	26.400%
2024	**OOS**	**−9.200%**	**−1.350**	**10.000%**	**−11.400%**	**24.900%**
2025	**OOS**	**12.800%**	**0.840**	**10.000%**	**−6.700%**	**17.900%**
2026	**OOS**	**31.500%**	**2.136**	**10.000%**	**−4.500%**	**15.100%**

Table 7: Annual performance summary under 10% target volatility scaling. IS = in-sample (2015–2023), OOS = out-of-sample (2024–2026).

Overall, the out-of-sample results show that the portfolio does not perform equally well across all regimes, but they do demonstrate persistence of edge, stability of the diversification structure, and positive expectancy on unseen data. The out-of-sample period captures both the cost of the portfolio's regime dependence (2024) and the benefit of its structural resilience (the 2025–2026 recovery).

7 Risks

Several important risks remain despite the portfolio's decent in-sample and out-of-sample performance.

The largest methodological risk is overfitting. The research process explored a broad space of signal families, conditioning filters, rebalance schedules, scaling rules, and portfolio combinations. Although the workflow emphasizes economic intuition and robustness rather than aggressive parameter tuning repeated experimentation inevitably increases the probability that some results reflect noise rather than persistent market structure. The out-of-sample period test mitigates this somewhat, but imperfectly.

The portfolio is clearly regime-dependent. Most sleeves are fundamentally mean-reversion or relative-value strategies. These sleeves can struggle during persistent directional trends, momentum-driven markets, or environments where convergence relationships break down. This behavior is visible during the 2024 out-of-sample period. The momentum sleeve partially offsets this exposure, but the portfolio still retains a structural bias toward convergence behavior.

Finally, implementation and capacity risks are meaningful. Several sleeves — particularly the always-on distance-pairs sleeve and the residual gap-reversion sleeve — are highly sensitive to transaction costs, slippage, and execution quality. The constant 10 bps transaction cost assumption is a simplification and may understate costs during stressed or crowded conditions. As capital scales, crowding effects, spread widening, borrow constraints, and execution delays could materially reduce realized performance even if the underlying signals remain statistically valid.

8 Conclusion

The final portfolio achieves a 2.039 net Sharpe ratio after 10 bps transaction costs through diversification across economically distinct signal families rather than reliance on any single dominant alpha source. Distance pairs relative value provides the portfolio's core while conditioned mean reversion, monotonic momentum, and gap reversion contribute complementary return streams with different regime sensitivities and low realized correlation. Conditioning filters materially improve implementation quality by reducing turnover and cost drag. The portfolio remains robust across weighting schemes, ablation tests, and elevated transaction cost assumptions. The out-of-sample results demonstrate regime sensitivity rather than structural collapse: performance weakens during the momentum dominated 2024 regime, but recovers through 2025 and 2026 while maintaining low benchmark correlation and controlled drawdowns.

References

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